

EEE4114F Class Test 2

May 2022

Duration: 1 hours

Total: 40 marks

Empl ID:

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Instructions:

- This is a closed-book test but you may use a calculator.
- There are two parts to this test.
- **Part A** has *two* questions totalling 20 marks. You must answer all of them.
- **Part B** has *two* questions totalling 20 marks. You must answer all of them.
- Part A and part B each count 50% of the final mark.
- You must use a pen to complete the exam — pencil will only be marked for drawings. Values indicated on drawings must be written in pen. Please show all of your working clearly — method and approach matter.
- A table of standard Fourier transform and z-transform pairs appears at the end of this paper.
- You have 1 hour.

Part A:	Question:	1	2	Total
	Points:	10	10	20
	Score:			

Part B:	Question:	3	4	Total
	Points:	10	10	20
	Score:			

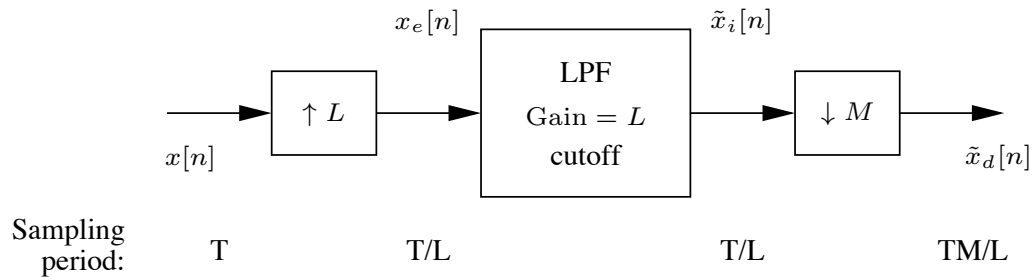
Part A: Digital Signal Processing

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Question 1 [10 marks]

Consider the system



where the cutoff of the LPF is at $\min(\pi/L, \pi/M)$. Suppose the input is

$$x[n] = \sin(2\pi n/3)/(\pi n),$$

with $L = 5$ and $M = 6$.

- Sketch the spectrum $X(e^{j\omega})$ of $x[n]$.
- Sketch the spectrum $X_e(e^{j\omega})$ of $x_e[n]$.
- Sketch the spectrum $\tilde{X}_i(e^{j\omega})$ of $\tilde{x}_i[n]$.
- Specify $\tilde{x}_d[n]$.

Solution:

- From tables we find that

$$X(e^{j\omega}) = \begin{cases} 1 & |\omega| \leq 2\pi/3 \\ 0 & 2\pi/3 < |\omega| \leq \pi, \end{cases}$$

and the sketch is given below.

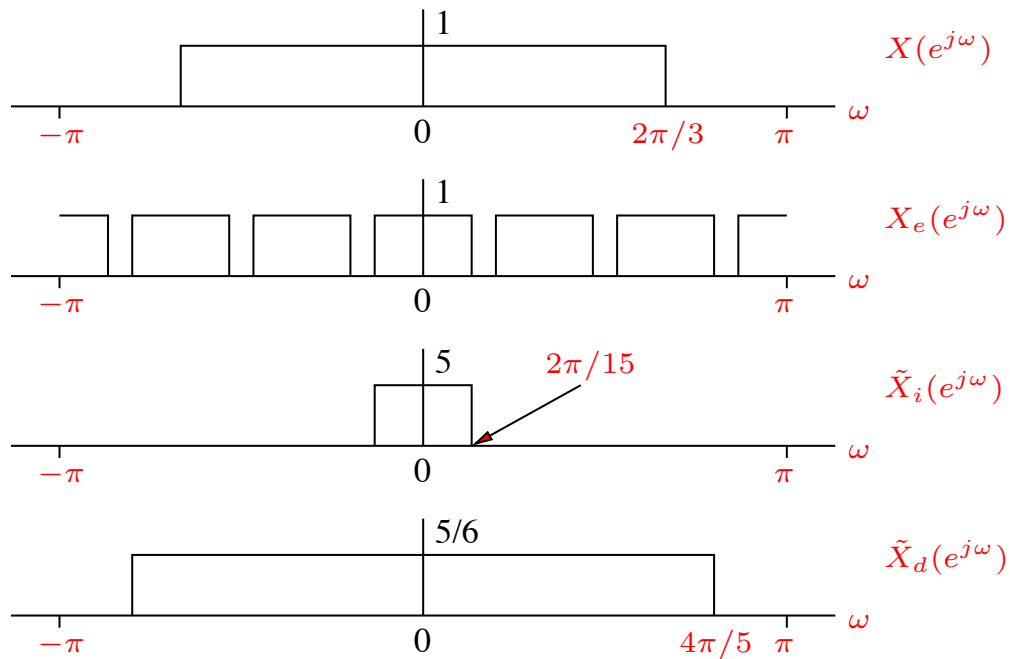
- The quantity $X_e(e^{j\omega}) = X(e^{j\omega L}) = X(e^{j5\omega})$. This is just the (periodic) function $X(e^{j\omega})$ compressed by a factor of 5 on the ω axis, and is sketched below.
- The ideal lowpass filter has a cutoff frequency of $\omega_c = \min(\pi/6, \pi/5) = \pi/6$ and a gain of 5. This filter just retains the central replica of $X_e(e^{j\omega})$ and scales it, and the output is sketched below.
- The downsampling stage expands the central replica by a factor of 6 and scales it by $1/6$. The spectrum of the output is sketched below, and can be expressed as

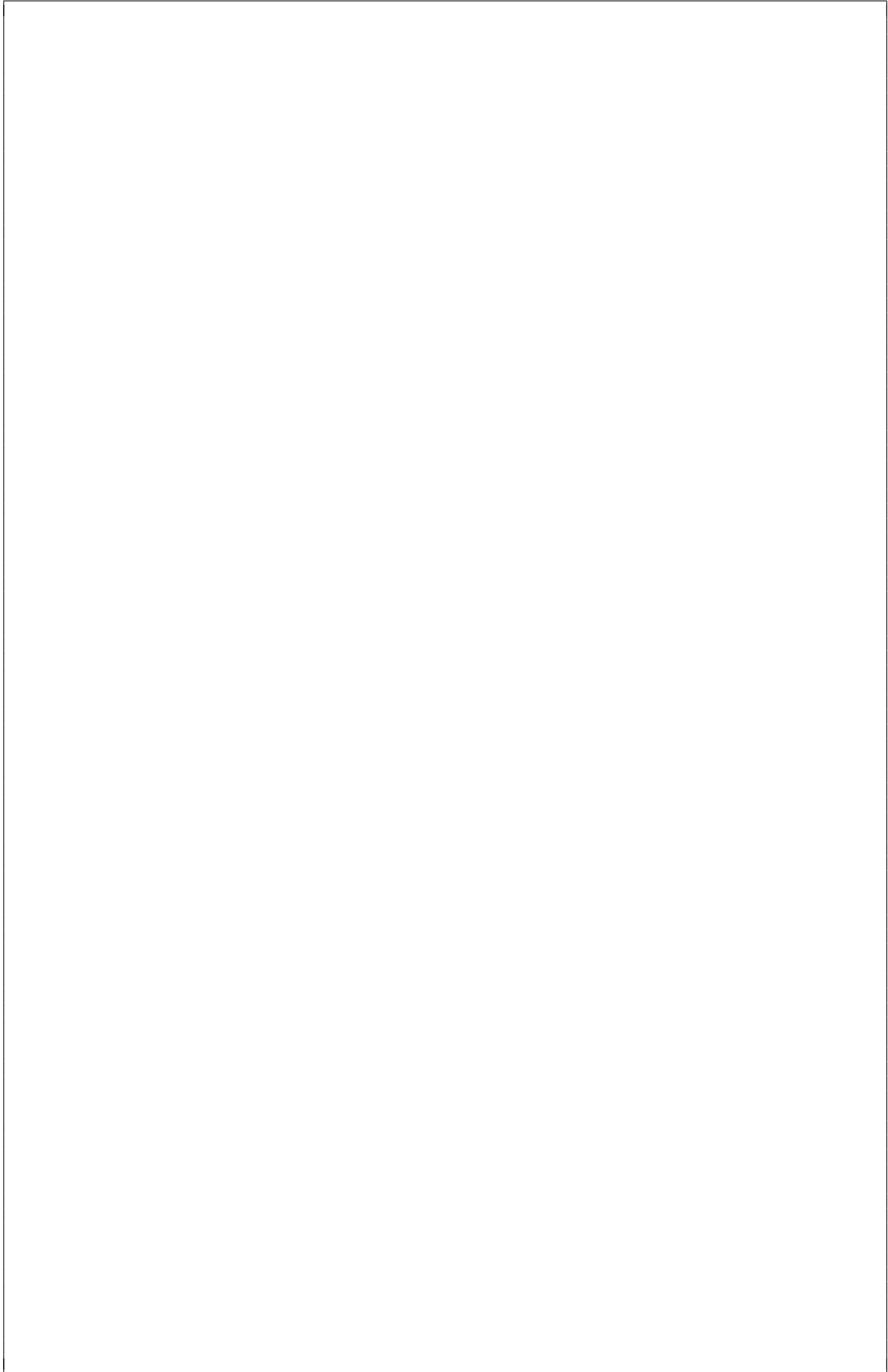
$$\tilde{X}_d(e^{j\omega}) = \begin{cases} 5/6 & |\omega| \leq 4\pi/5 \\ 0 & 4\pi/5 < |\omega| \leq \pi. \end{cases}$$

Thus

$$\tilde{x}_d[n] = 5/6 \frac{\sin(4\pi/5 n)}{\pi n}.$$

The sketches for all the answers given above are as follows. Note that all functions are periodic with period 2π and only the interval $-\pi$ to π are shown:





Question 2 [10 marks]

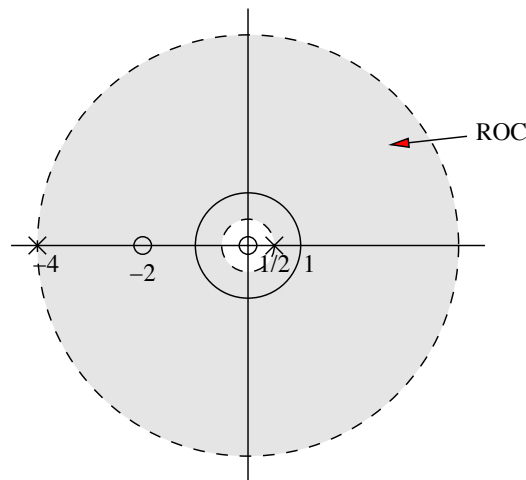
A stable linear time invariant system has the transfer function

$$H(z) = \frac{z(z+2)}{(z-1/2)(z+4)}.$$

1. Draw a pole-zero plot of the system function and indicate the region of convergence.
2. Find the frequency response $H(e^{j\omega})$ of this system.
3. Calculate the value of the frequency response $H(e^{j\omega})$ at $\omega = 0.2\pi$.
4. Find the output $y[n]$ produced by the input $x[n] = \cos(0.2\pi n)$. Express your result as real quantities (i.e. simplify to eliminate any instances of $j = \sqrt{-1}$).

Solution:

a. Pole-zero plot as follows:



b. The frequency response is

$$H(e^{j\omega}) = H(z)|_{z=e^{j\omega}} = \frac{e^{j\omega}(e^{j\omega} + 2)}{(e^{j\omega} - 1/2)(e^{j\omega} + 4)}.$$

c. The previous expression for $\omega = 0.2\pi$ gives

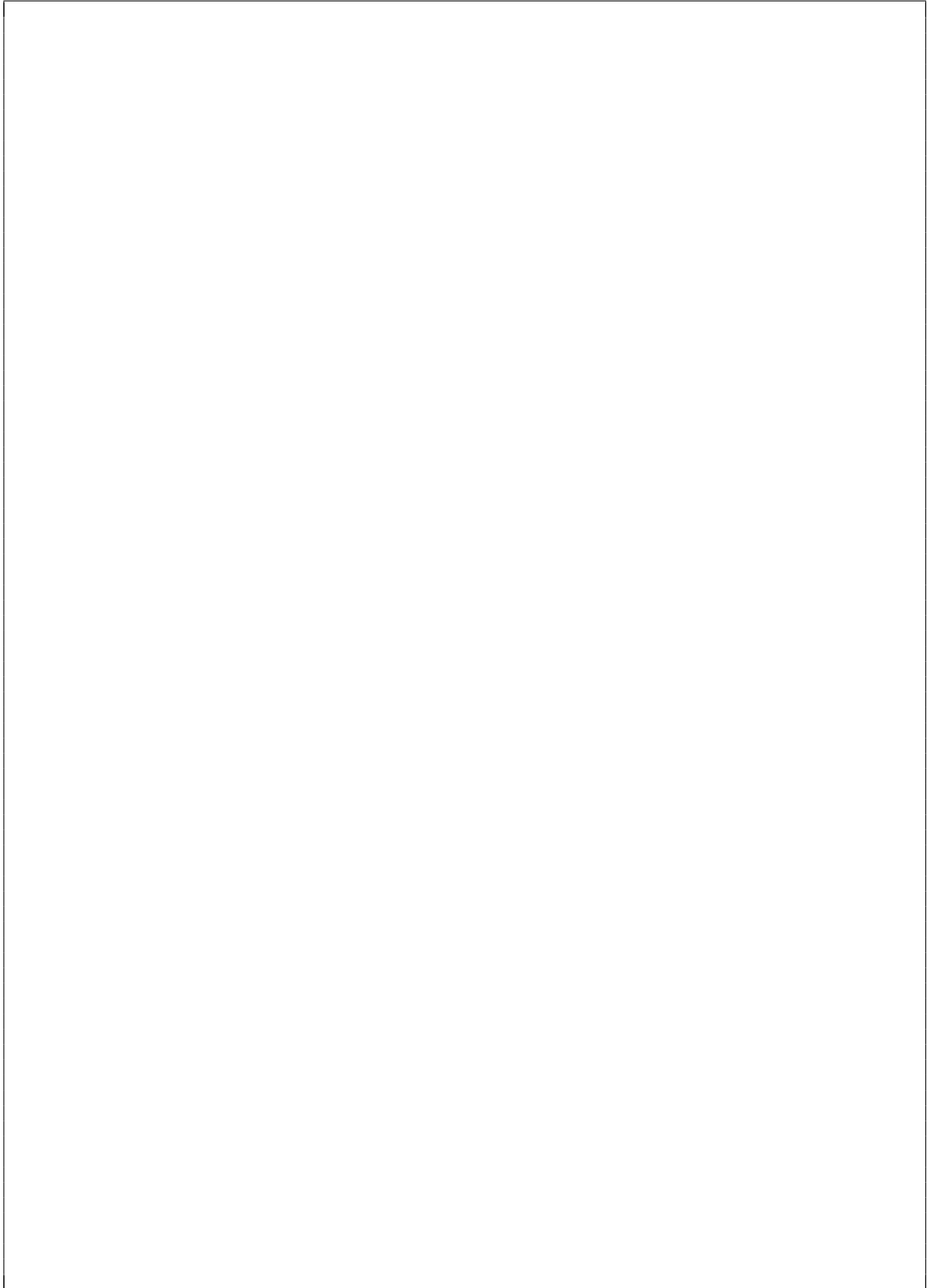
$$H(e^{j0.2\pi}) = \frac{e^{j0.2\pi}(e^{j0.2\pi} + 2)}{(e^{j0.2\pi} - 1/2)(e^{j0.2\pi} + 4)}.$$

This expression is good enough, although it can be evaluated to give $0.8920e^{-j0.3738}$.

d. Since the pole-zero plot is symmetric around the real axis, an input $\cos(\omega_0\pi n) = 1/2e^{j\omega_0 n} + 1/2e^{-j\omega_0 n}$ yields an output

$$\begin{aligned} y[n] &= 1/2H(e^{j\omega_0})e^{j\omega_0 n} + 1/2H(e^{-j\omega_0})e^{-j\omega_0 n} \\ &= 1/2|H(e^{j\omega_0})|e^{j(\omega_0 n + \angle H(e^{j\omega_0}))} + 1/2|H(e^{j\omega_0})|e^{-j(\omega_0 n + \angle H(e^{j\omega_0}))} \\ &= |H(e^{j\omega_0})| \cos(\omega_0 n + \angle H(e^{j\omega_0})). \end{aligned}$$

Evaluating for $\omega = 0.2\pi$ gives $y[n] = 0.8920 \cos(0.2\pi n - 0.3738)$.



END OF PART A

Part B: Machine Learning

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Question 3 [10 marks]

Suppose you are attempting to predict the temperature of some arbitrary device given the length of time it has been active for. You collect data and attempt to train three different models using a repeated holdout method to split the data into training and test sets. The outputs for all three models are shown in Figure 1 for just one split of the data.

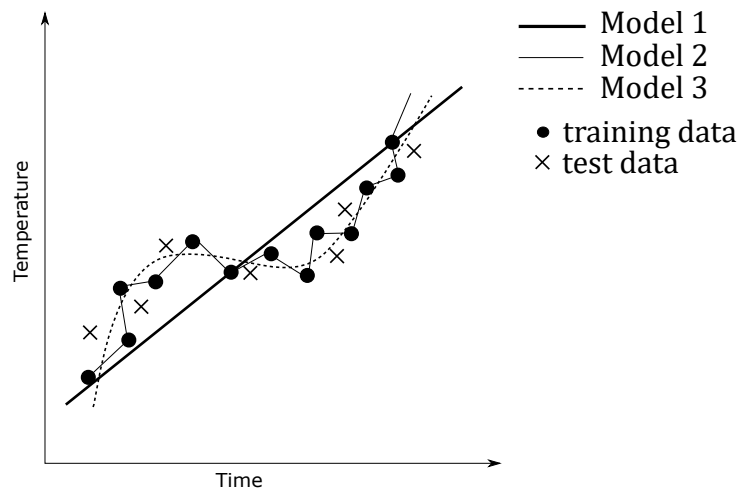


Figure 1: The output of three different models

- a. What is the main disadvantage of using repeated holdout compared to cross-validation? (2)

Solution: When using repeated holdout methods the test data across multiple splits are not independent, which can result in information leak. It is therefore possible that a model is trained and evaluated on many of the same samples, which results in a poor estimate of the model's variance.

You are given the box-and-whisker plots in Figure 2 for the three models based on the test errors recorded when using repeated holdout resampling of the data

- b. Which box-and-whisker plot (A, B and C) would most likely correspond to each model in Figure 1? Explain your answers. (6)

Solution: A would correspond to model 1 - high bias and variance, model has not fit well to the training data and does not appear to have the necessary complexity.
 B would correspond to model 3 - low bias and low variance, the model has fit well to the training data, and generalizes well to the test data as well.
 C would correspond to model 2 - low bias, but high variance. The model is overfit on the training data and does not generalize well to the test data.

- c. Which model would benefit the most from regularization? Explain your answer. (2)

Solution: Model 2 would benefit the most as it appears to have a large number of parameters based on its flexibility. Regularization tries to limit the effects of model parameters and often results in parameters being reduced to zero, or close to it. This effectively simplifies

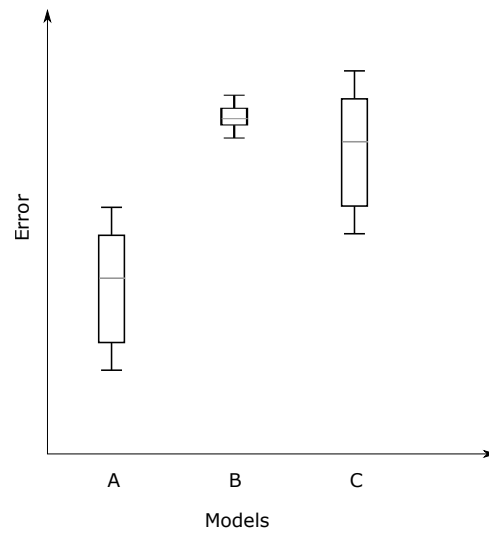
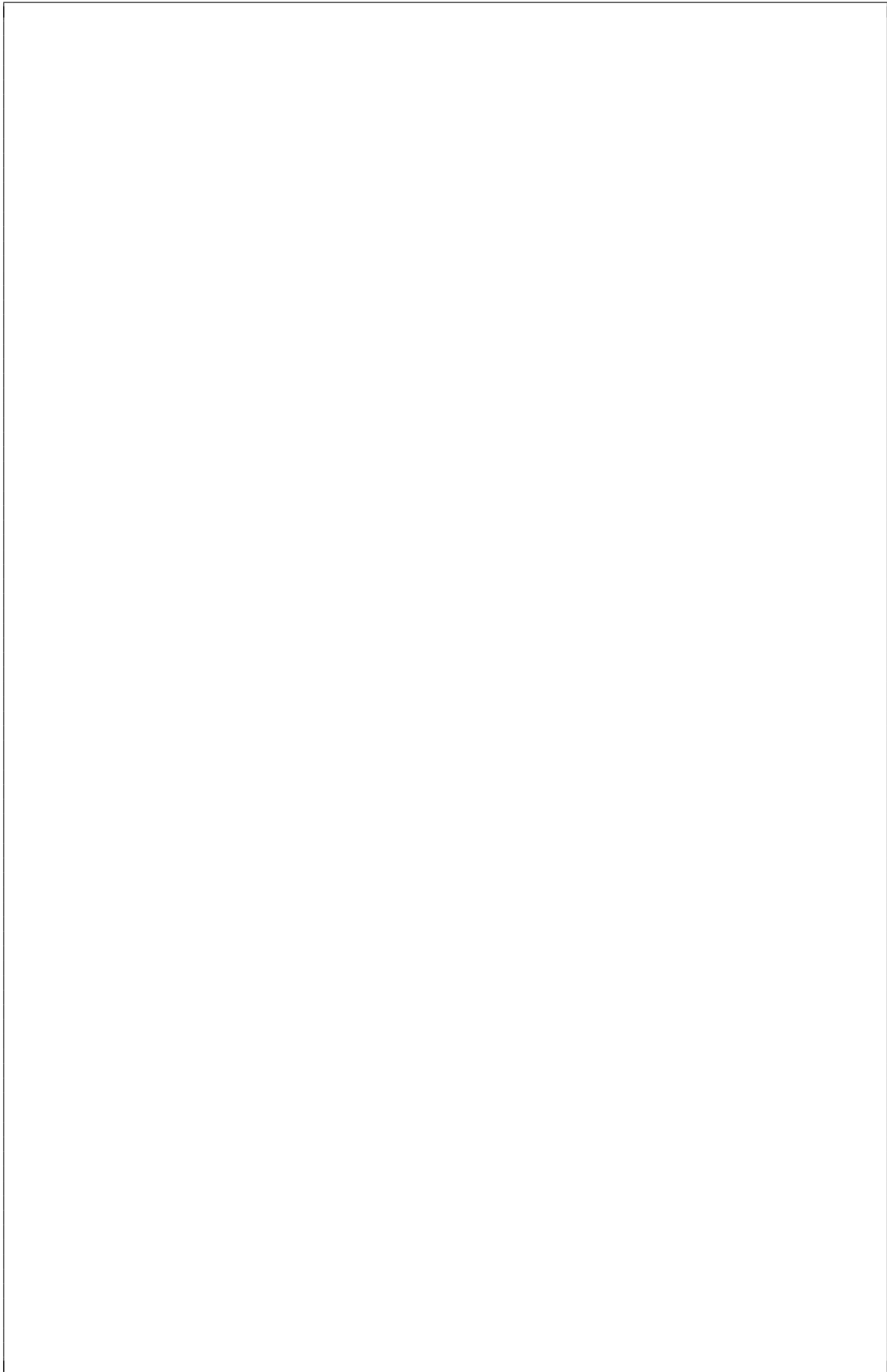


Figure 2: Box-and-whisker plots for the three models

the behaviour of the model, and in the case of model 2, could alleviate the clear overfitting problems that are shown.



Question 4 [10 marks]

Suppose you need to develop a system that predicts the concentration of SARS-CoV-2 based on the measurements collected from a CO_2 sensor. You collect a data set, $\mathcal{D}$, consisting of input-output pairs (x_i, y_i) where x_i is the CO_2 concentration, and y_i is the target variable, which in this case is the concentration of SARS-CoV-2.

$$\mathcal{D} = \{(1, 4), (2, 8), (3, 19), (4, 34)\}$$

You then decide to fit a regression model of the form $h_\theta(x) = \theta_0 + \theta_1 x^2$ to it using gradient descent optimization methods. The model's parameters are randomly initialised such that

$$\theta_0 = 0, \theta_1 = 1$$

. The objective function, J , you choose to optimize is given by the mean square error of the form:

$$J = \frac{1}{2N} \sum_{i=1}^N (h_\theta(x_i) - y_i)^2$$

Where N is the number of examples in a single training iteration.

Parameters are updated after **each** iteration and a single iteration uses one example from the data set . The update rule is given by:

$$\theta_i := \theta_i - \eta \frac{\partial J}{\partial \theta_i}$$

Where $\eta = 0.1$. Answer the following questions

- a. Write down the expressions for the gradients of the objective function with respect to the model's parameters $\frac{\partial J}{\partial \theta_0}$ and $\frac{\partial J}{\partial \theta_1}$. (2)

Solution:

$$\begin{aligned} \frac{\partial J}{\partial \theta_0} &= \frac{2}{2} \left(\sum_{i=1}^1 ((\theta_0 + \theta_1 x_i^2) - y_i) \right) = (\theta_0 + \theta_1 x_i^2) - y_i \\ \frac{\partial J}{\partial \theta_1} &= \frac{2}{2} \left(\sum_{i=1}^1 ((\theta_0 + \theta_1 x_i^2) - y_i) x_i^2 \right) = ((\theta_0 + \theta_1 x_i^2) - y_i) x_i^2 \end{aligned}$$

- b. Calculate the gradients of the parameters after one training iteration using the first training pair in the data set. (2)

Solution:

$$\begin{aligned} h_\theta(1) &= 0 + 1(1^2) = 1 \\ \frac{\partial J}{\partial \theta_0} &= (1 - 4) = -3 \\ \frac{\partial J}{\partial \theta_1} &= (1 - 4)(1) = -3 \end{aligned}$$

- c. Calculate the **model** parameters after the second training iteration. (4)

Solution: First we need to know the parameters after the first update.

$$\begin{aligned}\theta_0 &:= \theta_0 - \eta \frac{\partial J}{\partial \theta_0} \\ &:= 0 - 0.1(-3) \\ &:= 0.3\end{aligned}$$

$$\begin{aligned}\theta_1 &:= \theta_1 - \eta \frac{\partial J}{\partial \theta_1} \\ &:= 1 - 0.1(-3) \\ &:= 1.3\end{aligned}$$

Then repeat the calculation of gradients, but using the new parameters and new data $h_\theta(x_i) = 0.3 + 1.3(x_i^2)$. Where the next example is $(2, 8)$

$$h_\theta(2) = 0.3 + 1.3(2^2) = 5.5$$

$$\frac{\partial J}{\partial \theta_0} = (5.5 - 8) = -2.5$$

$$\frac{\partial J}{\partial \theta_1} = (5.5 - 8)(2^2) = -10$$

Then the new parameters are

$$\begin{aligned}\theta_0 &:= \theta_0 - \eta \frac{\partial J}{\partial \theta_0} \\ &:= 0.3 - 0.1(-2.5) \\ &:= 0.55\end{aligned}$$

$$\begin{aligned}\theta_1 &:= \theta_1 - \eta \frac{\partial J}{\partial \theta_1} \\ &:= 1.3 - 0.1(-10) \\ &:= 2.3\end{aligned}$$

Suppose you now try to train a fully-connected neural network with one hidden layer containing two neurons to solve the problem. The weights are initialised to $W = \begin{bmatrix} 2 & -1 \end{bmatrix}$ and the bias values are initialised to 1. The activation functions for each neuron are Rectified Linear Units (ReLU)

- d. Calculate the activation values for the hidden neurons given that the input is $x = 3$ (2)

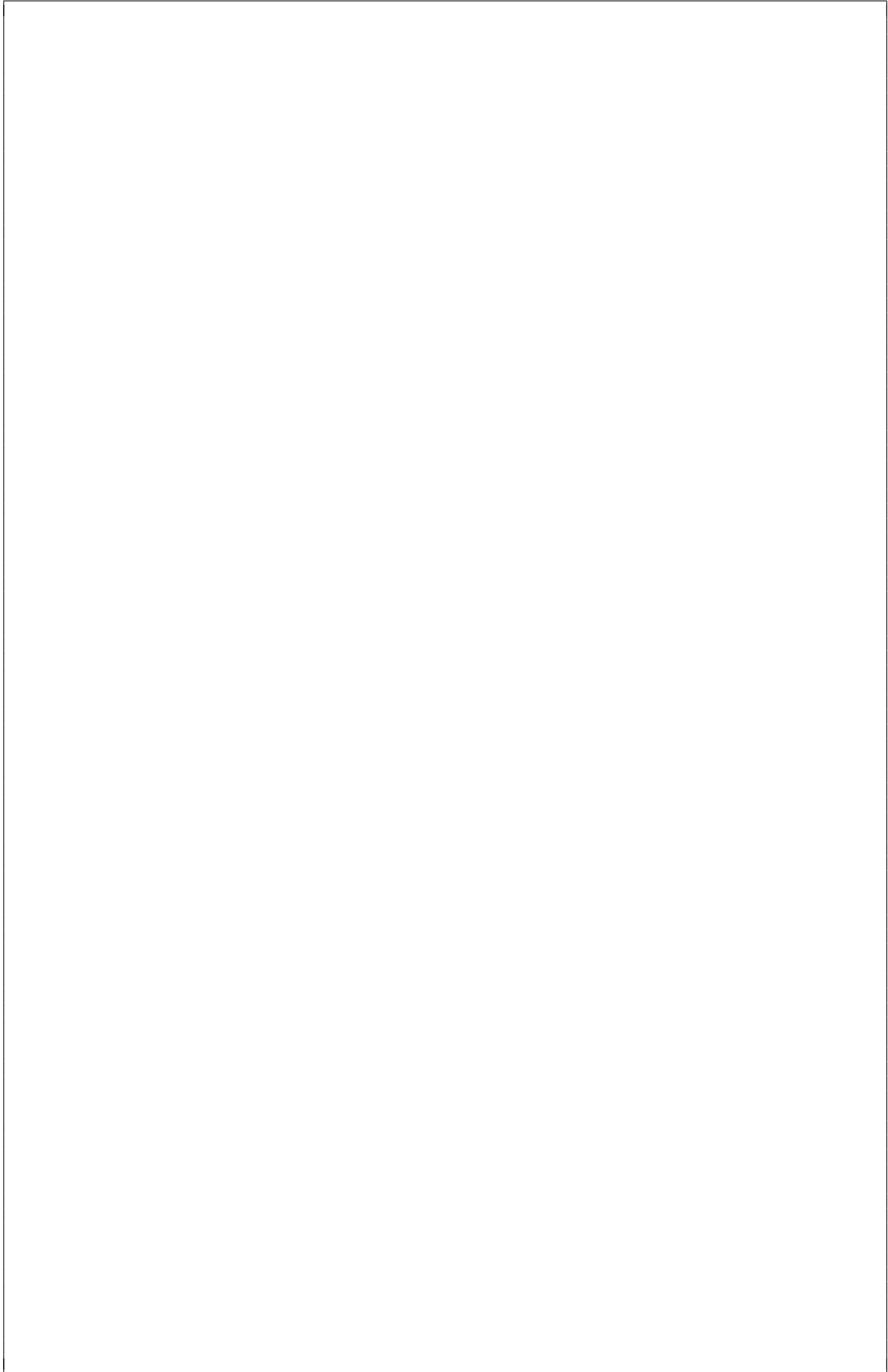
Solution:

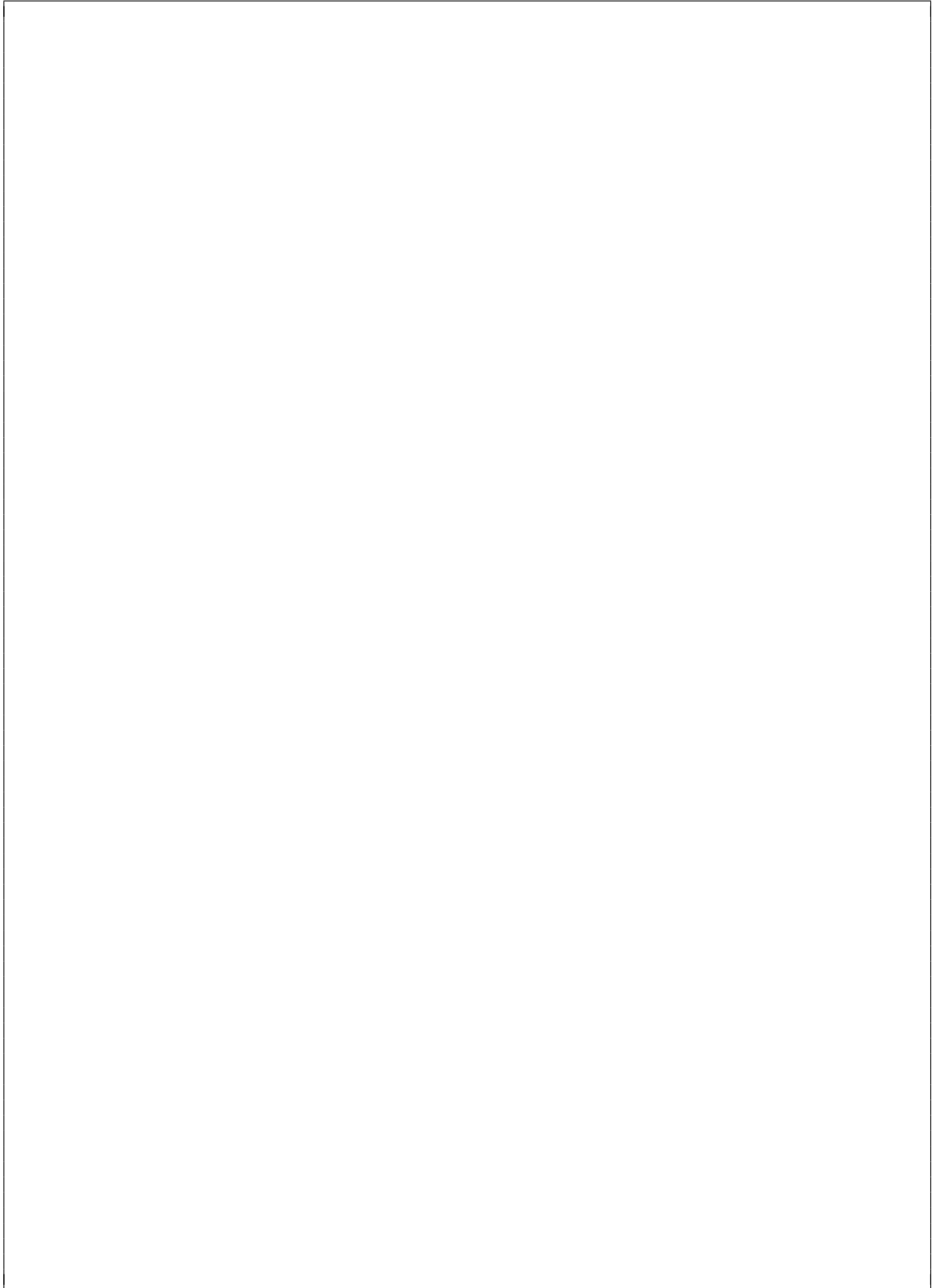
$$z_1 = w_1x + b_1 = 2(3) + 1 = 7$$

$$a_1 = \text{ReLU}(7) = 7$$

$$z_2 = w_2x + b_2 = -1(3) + 1 = -2$$

$$a_2 = \text{ReLU}(-2) = 0$$





END OF PART B

DSP information sheet

Transforms

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt \quad \Longleftrightarrow \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega \quad (\text{CTFT})$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega \quad (\text{DTFT})$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz \quad (\text{ZT})$$

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \quad \Longleftrightarrow \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with } W_N = e^{-j\frac{2\pi}{N}} \quad (\text{DFT})$$

Discrete-time Fourier transform properties

Sequences $x[n], y[n]$	Transforms $X(e^{j\omega}), Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d} X(e^{j\omega})$	Time shift
$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n+1)a^n u[n] \quad (a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \omega < \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n], y[n]$	Transforms $X(z), Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d} X(z)$	ROC = R_x	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC = $ z_0 R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC = $\frac{1}{R_x}$	Time reversal
$nx[n]$	$-z \frac{dX(z)}{dz}$	ROC = R_x	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC = R_x	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$ z > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$ z < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
$\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$ z > 0$
$\cos(\omega_0 n) u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$ z > 1$
$r^n \cos(\omega_0 n) u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$ z > r $

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for } a \neq 1.$$